Further Modelling with Functions

Exam code: 9MA0

Download PDF

Written by: Paul
Updated on 19 June 2025

SaveMyExams

Did this video help you? Yes No

Further modelling with functions

Making assumptions, criticisms and refinements of a model

- **Assumptions**, **criticisms** and **refinements** to models often need to be considered
- Real-life general knowledge and common sense can really help in mathematical modelling

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

... MAKE THE MATHEMATICS EASIER

CRITICISMS...

... ARE WAYS IN WHICH THE MODEL IS NOT SUITABLE

... LOOK FOR (EXTREME) VALUES WHERE THE MODEL BECOMES UNREALISTIC

REFINEMENTS...

... IMPROVE THE MODEL

... MAKE PREDICTED VALUES MORE REALISTIC

Copyright © Save My Exams. All Rights Reserved
save my exams

How do I use algebra to explore mathematical models?

- There will be no one-size-fits-all step-by-step guide to solving modelling questions
- A combination of skills and **problem-solving** skills will be needed

BOYLE'S IDEAL GAS LAW STATES THAT, FOR A CONSTANT TEMPERATURE, THE PRESSURE OF A GAS IS INVERSELY PROPORTIONAL TO THE VOLUME OF ITS CONTAINER.
A COMPANY IS TO USE THIS AS A MODEL TO FIND A SUITABLE CONTAINER FOR A PARTICULAR GAS THEY NEED TO STORE.

a) USING P FOR PRESSURE (MEASURED IN Pa, PASCALS) AND V FOR VOLUME (MEASURED IN m^3, CUBIC METRES) WRITE DOWN AN EQUATION FOR THE MODEL TO BE USED.

b) IT IS KNOWN THAT A CONTAINER WITH A VOLUME OF $5m^3$ CREATES A PRESSURE OF 350 000 Pa. FIND THE VALUE OF THE CONSTANT OF PROPORTIONALITY

c) FIND THE PRESSURE IF THE VOLUME OF THE CONTAINER IS 750 m^3

d) GIVEN THAT THE COMPANY MUST KEEP THE PRESSURE BELOW 120 000 Pa FIND, TO THE NEAREST CUBIC METRE, THE SMALLEST VOLUME OF CONTAINER THEY COULD USE

e) STATE ONE ASSUMPTION AND ONE PROBLEM IN USING THIS MODEL

Copyright © Save My Exams. All Rights Reserved
save my exams

b) SUBSTITUTE P AND V INTO MODEL EQUATION

$350\,000 = \frac{k}{5}$

$k = 1\,750\,000$

CAREFUL WITH ALL THOSE ZEROS

STORE IN CALCULATORS' MEMORY

c) $P = \frac{1\,750\,000}{V}$

$P = \frac{1\,750\,000}{750}$

$P = 2\,330$ Pa (3sf)

d) $P < 120\,000$

$\frac{1\,750\,000}{V} < 120\,000$

$V > \frac{1\,750\,000}{120\,000}$

$V > 14.583...$

∴ SMALLEST CONTAINER IS $15\,m^3$

e) ASSUMPTION:
THE AMOUNT OF GAS THE COMPANY NEEDS TO STORE DOES NOT CHANGE

PROBLEM:
KEEPING THE TEMPERATURE OF THE GAS/CONTAINER CONSTANT

Copyright © Save My Exams. All Rights Reserved
save my exams

Examiner Tips and Tricks

- Reciprocal graphs generally have two parts/curves
  - Only one – usually the positive – may be relevant to the model
  - Think about why **x/t/θ** can only take positive values?

Worked Example

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.